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Amended Claims

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MAIN REQUEST

CLAIMS

184. A bioactive composition comprising:
an isolated bioactive fraction derived from a *Theacea-Camellia* plant produced according to the method of any of claims 1-16, wherein said bioactive fraction is selected from the group consisting of a cell walls fraction, a cell walls fraction extract, a membrane fraction, a membrane fraction extract, a cytoplasm fraction, a cytoplasm fraction extract, a cell juice serum, and combinations thereof.

~~2. The bioactive composition according to claim 1, wherein said bioactive fraction is a cell walls fraction.~~

~~3. The bioactive composition according to claim 1, wherein said bioactive fraction is a cell walls fraction extract.~~

~~4. The bioactive composition according to claim 3, wherein said cell walls fraction extract has a total catechin content of between about 2.1 and about 4.5 milligrams per gram of dry matter.~~

~~5. The bioactive composition according to claim 3, wherein said cell walls fraction extract has a catechin content profile comprising:~~

~~between about 2.0 and about 3.0 milligrams of (+) catechin per gram of dry matter of the cell walls fraction extract,~~

~~between about 0.005 and about 0.02 milligrams of (-) epicatechin per gram of dry matter of the cell walls fraction extract,~~

~~between about 0.005 and about 0.02 milligrams of (-) epigallocatechin gallate per gram of dry matter of the cell walls fraction extract, and~~

~~between about 0.003 and about 0.01 milligrams of (-) epicatechin gallate per gram of dry matter of the cell walls fraction extract.~~

~~6. The bioactive composition according to claim 1, wherein said bioactive fraction is a membrane fraction.~~

~~7. The bioactive composition according to claim 1, wherein said bioactive fraction is a membrane fraction extract.~~

138. The method bioactive composition according to claim 37, wherein said membrane

fraction extract has a total catechin content of between about 15.0 and about 30.5 milligrams per gram of dry matter.

~~149.~~ The method ~~bioactive composition~~ according to claim ~~37~~, wherein said membrane fraction extract has a catechin content profile comprising:

between about 1.7 and about 3.3 milligrams of (-)-epigallocatechin per gram of dry matter of the membrane fraction extract,

between about 6.1 and about 10.2 milligrams of (+)-catechin per gram of dry matter of the membrane fraction extract,

between about 0.3 and about 1.1 milligrams of (-)-epicatechin per gram of dry matter of the membrane fraction extract,

between about 6.2 and about 12.5 milligrams of (-)-epigallocatechin gallate per gram of dry matter of the membrane fraction extract,

between about 0.007 and about 0.03 milligrams of (-)-gallocatechin gallate per gram of dry matter of the membrane fraction extract, and

between about 1.3 and about 3.3 milligrams of (-)-epicatechin gallate per gram of dry matter of the membrane fraction extract.

~~10.~~ The ~~bioactive composition~~ according to claim ~~1~~, wherein said ~~bioactive fraction~~ is a ~~cytoplasm fraction~~.

~~11.~~ The ~~bioactive composition~~ according to claim ~~1~~, wherein said ~~bioactive fraction~~ is a ~~cytoplasm fraction extract~~.

~~12.~~ The ~~bioactive composition~~ according to claim ~~1~~, wherein said ~~bioactive fraction~~ is a ~~cell juice serum~~.

~~1513.~~ The method ~~bioactive composition~~ according to claim ~~612~~, wherein said cell juice serum has a total catechin content of between about 8.0 and about 20.0 milligrams per gram of dry matter.

~~1614.~~ The method ~~bioactive composition~~ according to claim ~~612~~, wherein said cell juice serum has a catechin content profile comprising:

between about 2.1 and about 4.4 milligrams of (-)-epigallocatechin per gram of dry matter of the cell juice serum,

between about 4.2 and about 8.6 milligrams of (+)-catechin per gram of dry matter of the cell juice serum,

between about 0.2 and about 2.0 milligrams of (-)-epicatechin per gram of dry matter of the cell juice serum,

between about 1.2 and about 3.2 milligrams of (-)-epigallocatechin gallate per gram of dry matter of the cell juice serum,

between about 0.01 and about 0.1 milligrams of (-)-gallocatechin gallate per gram of dry matter of the cell juice serum, and

between about 0.2 and about 1.3 milligrams of (-)-epicatechin gallate per gram of dry matter of the cell juice serum.

1945. The bioactive composition according to claim 184 further comprising a stabilizing agent.

2046. The bioactive composition according to claim 1845, wherein said stabilizing agent is selected from the group consisting of an emulsifier, a preservative, an antioxidant, a polymer matrix, and mixtures thereof.

2147. A bioactive topical formulation suitable for topical application to a mammal, said bioactive topical formulation comprising:

a topically effective amount of the bioactive composition according to claim 184 and

a topically acceptable carrier.

2248. The bioactive topical formulation according to claim 2147, wherein the topically acceptable carrier is selected from the group consisting of a hydrophilic cream base, a hydrophilic lotion base, a hydrophilic surfactant base, a hydrophilic gel base, a hydrophilic solution base, a hydrophobic cream base, a hydrophobic lotion base, a hydrophobic surfactant base, a hydrophobic gel base, and a hydrophobic solution base.

2349. The bioactive topical formulation according to claim 2147, wherein the bioactive composition is present in an amount ranging from between about 0.001 percent and about 90 percent of the total weight of the bioactive topical formulation.

2420. Use of a bioactive composition according to any of claims 184 to 2049 or a bioactive topical formulation according to any of claims 21 to 23 in the manufacture of a medicament for inhibiting inflammatory activity in skin tissue of a mammal by applying the medicament bioactive composition to the skin tissue in an amount effective to inhibit inflammatory activity in the skin tissue.

2524. Use of a bioactive composition according to any of claims 184 to 2049 or a bioactive topical formulation according to any of claims 21 to 23 in the manufacture of a medicament for protecting skin tissue of a mammal from ultraviolet light-induced damage by applying the medicament bioactive composition to the skin tissue in an amount effective to reduce ultraviolet light induced damage of the skin tissue and to prevent oxidative damage of the skin tissue.

2622. The use according to claim 2524, wherein said ultraviolet light-induced damage is caused by ultraviolet light in a range of between about 320 and about 400 nanometers.

2723. Use of a bioactive composition according to any of claims 184 to 2049 or a bioactive topical formulation according to any of claims 21 to 23 in the manufacture of a medicament for normalizing skin disorders in skin tissue of a mammal, by applying the medicament bioactive composition to the skin tissue in an amount effective to normalize a cell disorder in the skin tissue.

28. A bioactive composition according to any of claims 18 to 20 or a bioactive topical formulation according to any of claims 21 to 23 for use in inhibiting inflammatory activity in skin tissue of a mammal, or for use in protecting skin tissue of a mammal from ultraviolet light-induced damage, or for use in normalizing skin disorders in skin tissue of a mammal, by applying the bioactive composition or topical formulation to the skin tissue of the mammal.

24. The use according to any of claims 20, 21 or 23 wherein said *Theacea* plant is a *Camellia* plant or a *Eurya* plant.

25. The use according to any of claims 20, 21 or 23, wherein said bioactive composition further comprises a stabilizing agent.

26. The use according to claim 25, wherein said stabilizing agent is selected from the group consisting of an emulsifier, a preservative, an antioxidant, a polymer matrix, and mixtures thereof.

27. The use according to any of claims 20, 21 or 23, wherein said bioactive composition further comprises a topically acceptable carrier.

~~28. The use according to claim 27, wherein the topically acceptable carrier is selected from the group consisting of a hydrophilic cream base, a hydrophilic lotion base, a hydrophilic surfactant base, a hydrophilic gel base, a hydrophilic solution base, a hydrophobic cream base, a hydrophobic lotion base, a hydrophobic surfactant base, a hydrophobic gel base, and a hydrophobic solution base.~~

129. A method for isolating a bioactive fraction derived from cell juice of a *Theaceae Camellia* plant, said method comprising:

providing a *Theaceae* plant fresh biomass of *Camellia* plant, wherein said fresh biomass comprises fresh *Camellia* plant biomass that has not undergone conventional tea processing;

separating the *Theaceae* plant fresh biomass of the *Camellia* plant into cell juice and a cell walls component, wherein said separating comprises grinding, macerating, and pressing the fresh biomass to separate the fresh biomass into the cell juice and the cell walls component;

treating the cell juice under conditions effective to yield a bioactive fraction, said conditions comprising subjecting the cell juice to microwave treatment to induce coagulation, thereby yielding a coagulated membrane fraction and a non-coagulated portion of the cell juice, wherein said bioactive fraction is selected from the group consisting of a membrane fraction, a membrane fraction extract, a cytoplasm fraction, a cytoplasm fraction extract, and a cell juice serum; and

isolating said bioactive fraction from the treated cell juice.

230. The method according to claim 129, wherein said bioactive fraction is a membrane fraction.

334. The method according to claim 129, wherein said bioactive fraction is a membrane fraction extract.

432. The method according to claim 129, wherein said bioactive fraction is a cytoplasm fraction.

533. The method according to claim 129, wherein said bioactive fraction is a cytoplasm fraction extract.

634. The method according to claim 129, wherein said bioactive fraction is a cell juice serum.

7. The method according to claim 1, wherein said *Camellia* plant is selected from the group consisting of *Camellia sinensis*, *Camellia japonica*, *Camellia reticulata*, and *Camellia sasanqua*.
8. The method according to claim 1, wherein said isolating comprises:
separating the coagulated membrane fraction from the non-coagulated portion,
thereby yielding the membrane fraction.
9. The method according to claim 8, wherein the membrane fraction is subjected to solvent extraction with dimethyl sulfoxide to yield the membrane fraction extract.
10. The method according to claim 1, wherein said isolating comprises:
separating the coagulated membrane fraction from the non-coagulated portion,
thereby yielding the non-coagulated portion; and
subjecting the non-coagulated portion to coagulation by isoelectric precipitation and centrifugation, thereby yielding a precipitate comprising the cytoplasm fraction.
11. The method according to claim 10, wherein the cytoplasm fraction is subjected to solvent extraction with dimethyl sulfoxide to yield the cytoplasm fraction extract.
12. The method according to claim 1, wherein said isolating comprises:
separating the coagulated membrane fraction from the non-coagulated portion,
thereby yielding the non-coagulated portion; and
subjecting the non-coagulated portion to coagulation by isoelectric precipitation and centrifugation, thereby yielding a supernatant liquid comprising the cell juice serum.
1735. An isolated bioactive fraction derived from a *Camellia* plant composition-produced according to the method of any of claims 1-1629.
2936. A method for isolating a bioactive fraction derived from a cell walls component of a *Camellia Theaceae*-plant, said method comprising:
providing a *Camellia Theaceae*-plant;
separating the *Camellia Theaceae*-plant into cell juice and a cell walls component;
treating the cell walls component under conditions effective to yield a bioactive fraction; and
isolating the bioactive fraction from the treated cell walls component.

| 3037. The method according to claim 2936, wherein said bioactive fraction is a cell walls fraction.

| 3138. The method according to claim 2936, wherein said bioactive fraction is a cell walls fraction extract.

| ~~39. The method according to claim 29 or 36, or bioactive composition according to Claim 1, wherein said *Theacea* plant is a *Camellia* plant or a *Eurya* plant.~~

| 3240. The method or bioactive composition according to claim 2939, wherein said *Camellia* plant is selected from the group consisting of *Camellia sinensis*, *Camellia japonica*, *Camellia reticulate*, and *Camellia sasanqua*.

| ~~41. The method or bioactive composition according to claim 39, wherein said *Eurya* plant is *Eurya sandwicensis*.~~

| 3342. An isolated bioactive fraction produced according to the method of claim 2936.